Beom Park

Ph.D. Candidate
School of Aeronautics and Astronautics
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Research Interests

Multi-body dynamics; Dynamical systems theory; Spacecraft trajectory design and optimization;

Education

2021-2025 (expected)	**Ph.D. Candidate** in Aeronautics and Astronautics, Purdue University, U.S. Advisor: Professor Kathleen Howell Thesis Title: Transitioning from a Lower- to a Higher-Fidelity Model: Leveraging an Intermediate Model in the Earth-Moon L1/L2 Regions, *in preparation* Committee: Professor Carolin Frueh, Professor Kenshiro Oguri, Dr. Diane Davis
2018-2021	**M.S.** in Aeronautics and Astronautics, Purdue University, U.S. Advisor: Professor Kathleen Howell Thesis Title: Low-Thrust Trajectory Design for Tours of the Martian Moons [link] Committee: Professor Carolin Frueh, Professor James Longuski
2012-2018	**B.S.**, College of Liberal Studies (Aerospace Engineering), Seoul National University, Republic of Korea

Honors and Awards

2025	*Magoon Research Excellence Award*: Awarded by College of Engineering, Purdue University
2025	*American Astronautical Society John V. Breakwell Student Award*
2024	*Korea Aerospace Research Institute's Travel Grant*: Awarded for participating in the International Astronautical Congress 2024, Milan, Italy
2021-2025	*Kwanjeong Scholarship*: Awarded by Kwanjeong Educational Foundation for Ph.D. studies abroad
2018-2020	*Kwanjeong Scholarship*: Awarded by Kwanjeong Educational Foundation for M.S. studies abroad

2018	*Graduation with Honor*: Awarded by the Seoul National University Alumni Association for the academic excellence upon graduation

Research Projects

2024-2025	Lunar constellation design leveraging frozen orbits Sponsor: Intuitive Machines Relevant publications: C2.
2022-2025	Characterizing cislunar dynamical regime in a higher-fidelity model Partial sponsor: NASA Johnson Space Center Relevant publications: P1. J2. J3. C1. C3. C4. C5. C6. C8.
2021-2024	Trajectory design and analysis for the IM-1 (*first lunar lander from the private sector in human history*) & Khon-1 mission [link] Sponsor: Intuitive Machines Relevant publication: C7.
2021	Low-thrust trajectory design for the Lunar IceCube (LIC) mission [link] Sponsor: NASA Goddard Space Flight Center Relevant publication: C9.

Publications

Journal Papers

J1. Park, B., Sanaga, R. R., and Howell, K. (2025) "A Frequency-Based Hierarchy of Dynamical Models in Cislunar Space: Leveraging Periodically and Quasi-Periodically Perturbed Models," *Celestial Mechanics and Dynamical Astronomy*, 137, 5. [link]

J2. Park, B., and Howell, K. (2025) "Characterization of Earth-Moon L2 Halo Analogs in an Ephemeris Model Utilizing the Elliptic Restricted Three-Body Problem," *Advances in Space Research*, In press. [link]

J3. Park, B., and Howell, K. (2024) "Assessment of Dynamical Models for Transitioning from the Circular Restricted Three-Body Problem to an Ephemeris Model with Applications," *Celestial Mechanics and Dynamical Astronomy*, 136, 6. [link] - invited to a webinar[link]

J4. Canales, D., Gupta, M., Park, B., and Howell, K. (2022) "A Transfer Trajectory Framework for the Exploration of Phobos and Deimos Leveraging Resonant Orbits," *Acta Astronautica*, 194, 263-276. [link]

Conference Proceedings / National (International) Presentations

C1. Sanaga, R. R., Park, B., and Howell, K. C. (2025) "A Unified Numerical Transition Scheme between Dynamical Models within Cislunar Space," *35th AAS/AIAA Space Flight Mechanics Meeting*, Kaua'i, Hawaii, U.S., January 19-23, 2025. [link]

C2. Park, B., Howell, K. C., and Stewart, S. (2025) "Elliptical Lunar Frozen Orbit Constellation Design within a Model of Evolving Fidelity," *35th AAS/AIAA Space Flight Mechanics Meeting*, Kaua'i, Hawaii, U.S., January 19-23, 2025. [link]

C3. Park, B., Sanaga, R. R., and Howell, K. C. (2025) "Numerical Assessment of Intermediate Models in a Frequency-Based Hierarchy for the Cislunar Domain," *35th AAS/AIAA Space Flight Mechanics Meeting*, Kaua'i, Hawaii, U.S., January 19-23, 2025. [link] - awarded John V. Breakwell Student Award

C4. Park, B., Sanaga, R. R., and Howell, K. C. (2024) "Bridging Ephemeris Transition Gaps: Quasi-Periodic Extensions for the Hill Restricted Four-Body Problem in Cislunar Space," *75th International Astronautical Congress*, Milan, Italy, October 14-18, 2024. [link] - awarded Korea Aerospace Research Institute's Travel Grant

C5. Park, B. and Howell, K. C. (2024) "Characterizing Transition-Challenging Regions Leveraging the Elliptic Restricted Three-Body Problem: L2 Halo Orbits," *AIAA SciTech Forum*, Orlando, Florida, U.S., January 8-12, 2024. [link]

C6. Park, B. and Howell, K. C. (2023) "Leveraging the Elliptic Restricted Three-Body Problem for Characterization of Multi-Year Earth-Moon L2 Halos in an Ephemeris Model," *AAS/AIAA Astrodynamics Specialist Conference*, Big Sky, Montana, U.S., August 13-17, 2023. [link]

C7. Hoffman, A., Park, B., and Howell, K. C. (2022) "Trajectory Design for a Secondary Payload Within a Complex Gravitational Environment: The Khon-1 Spacecraft," *AAS/AIAA Astrodynamics Specialist Conference*, Charlotte, North Carolina, U.S., August 7-10, 2022. [link]

C8. Park, B. and Howell, K. C. (2022) "Leveraging Intermediate Dynamical Models for Transitioning from the Circular Restricted Three-Body Problem to an Ephemeris Model," *AAS/AIAA Astrodynamics Specialist Conference*, Charlotte, North Carolina, U.S., August 7-10, 2022. [link]

C9. Park, B., Howell, K. C., and Folta, D. C. (2021) "Design of Low-Thrust Transfers from an NRHO to Low Lunar Orbits: Applications for Small Spacecraft," *AAS/AIAA Astrodynamics Specialist Conference*, Big Sky, Montana (Virtual), U.S., August 9-11, 2021. [link]

C10. Canales, D., Gupta, M., Park, B., and Howell, K. C. (2021) "Exploration of Deimos and Phobos Leveraging Resonant Orbits," *31st AAS/AIAA Spaceflight Mechanics Meeting*, Charlotte, North Carolina (Virtual), U.S., February 1-3, 2021. [link]

Preprints

P1. Park, B., Sanaga, R. R., and Howell, K. (2025) "Bridging Ephemeris Transition Gaps: Quasi-Periodic Extensions for the Hill Restricted Four-Body Problem in Cislunar Space," *Acta Astronautica*, Under Review

P2. Park, B., Sanaga, R. R., and Howell, K. (2025) "Numerical Assessment of intermediate models in a Frequency-Based Hierarchy for The Cislunar domain," *Journal of Guidance, Control, and Dynamics*, Under Review.

P3. Sanaga, R. R., Park, B., and Howell, K. (2025) "A Unified Numerical Transition Scheme between Dynamical Models within Cislunar Space," *Journal of Astronautical Sciences*, Under Review

Talks

2024	Invited talk at *Celestial Mechanics and Dynamical Astronomy*'s webinar on J3.[link], selected "as being particularly innovative and/or having had significant recent impact"

Teaching Experiences

2020 **Teaching Assistant**, School of Aeronautics and Astronautics, Purdue University.
- AAE 440/590 (Spacecraft Attitude Dynamics)
- AAE 532 (Orbit Dynamics)

2019 **Teaching Assistant**, Department of Mathmatics, Purdue University.
- MA 161 (Plane Analytic Geometry And Calculus I)

Professional Experience

2013-2015 **Air Surveilance Operator (Sergeant)**, Republic of Korea Air Force

Academic Services

Journal of Spacecraft and Rockets (2 papers), Journal of Astronautical Sciences (3), Astrodynamics (1)